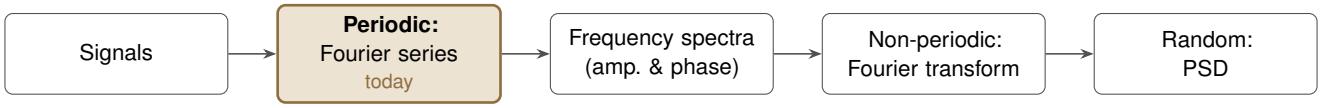
ME 365 · LECTURE 21

Spectral Analysis 1: Fourier Series

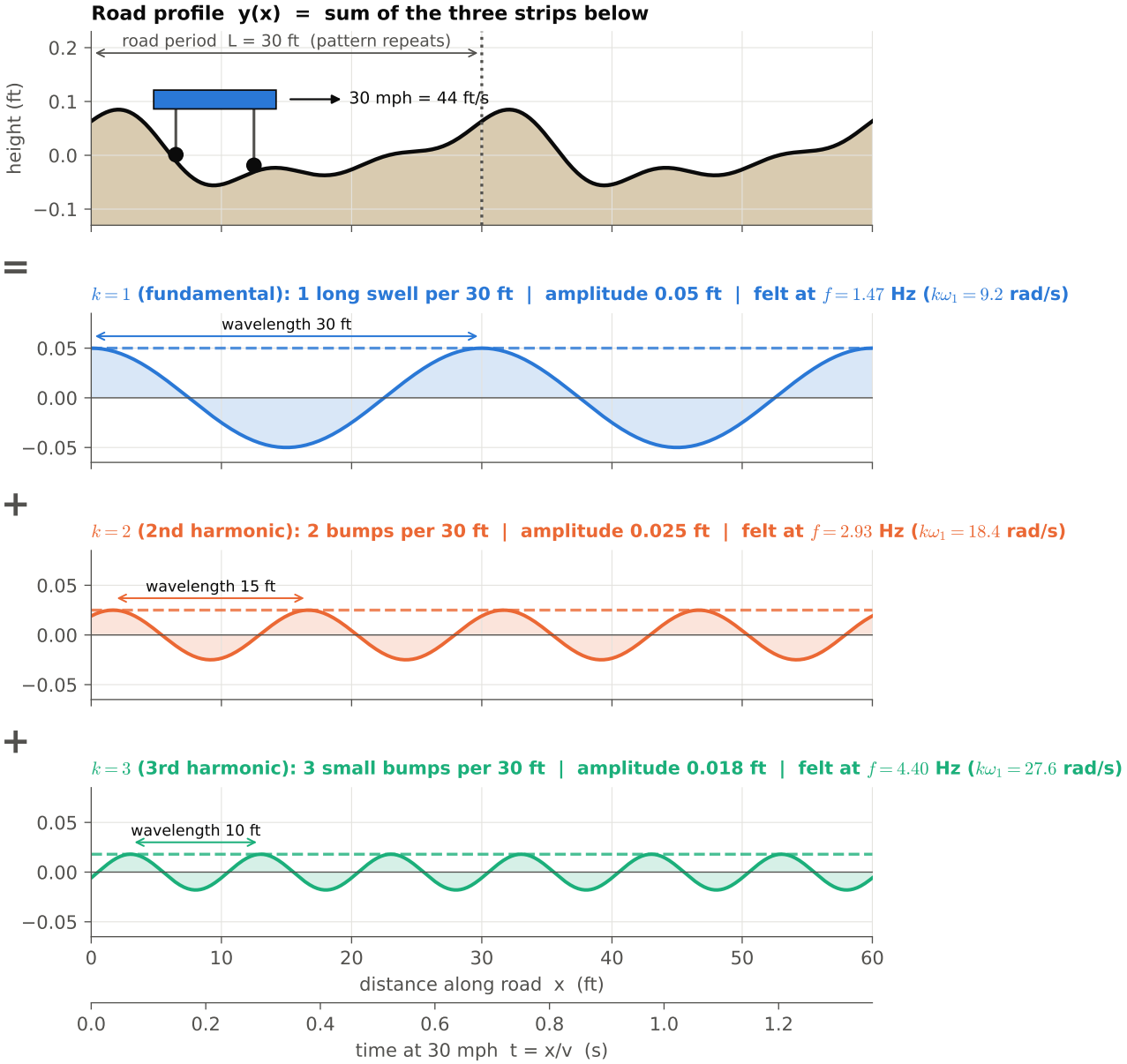


1 Why break signals into sinusoids?

We already know how an LTI system $G(s)$ responds to *one* sinusoid. A real input (such as a road profile) is not one sinusoid. Two facts let us handle it anyway:

- **Superposition:** $a\,u_1 + b\,u_2 \;\mapsto\; a\,y_1 + b\,y_2$.
- **Frequency response:** $M\cos(\omega t + \phi) \;\mapsto\; M\,|G(j\omega)|\cos\big(\omega t + \phi + \angle G(j\omega)\big)$ in steady state.

If $u(t) = \frac{A_0}{2} + \sum_k M_k \cos(k\omega_1 t - \theta_k)$, then the steady-state output is



The road (top) is exactly the sum of the three colored strips. Each strip is one *harmonic*: a perfectly sinusoidal road whose wavelength is L/k (30, 15, 10 ft). The car drives over each strip at 44 ft/s, so every strip shakes it at a single frequency $k\omega_1$, which is a sinusoidal input we already know how to handle with $G(j\omega)$. Vertical scale exaggerated.

How does a road wavelength turn into a frequency the car feels?

Signal type	Tool	Spectrum
Periodic	Fourier **series**	lines at $k\omega_1$ only
Non-periodic (well-defined)	Fourier **transform**	continuous in ω
Random	Power spectral density	continuous, statistical

2 The Fourier series

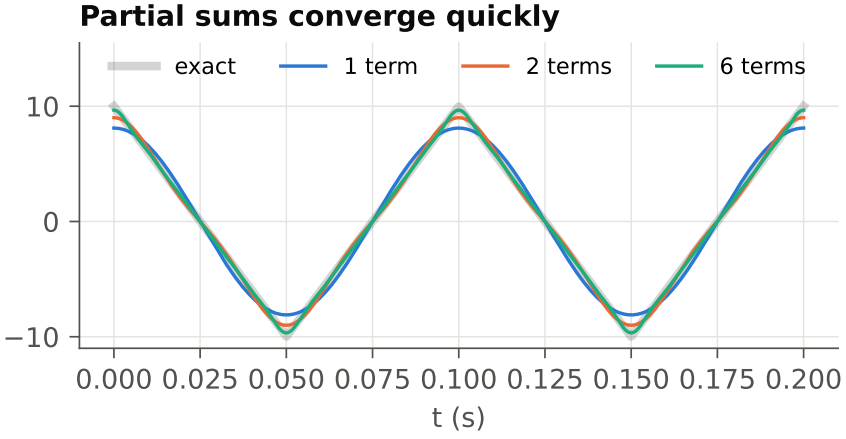
For $x(t+T) = x(t)$, with fundamental frequency $\omega_1 =$ _____ rad/s:

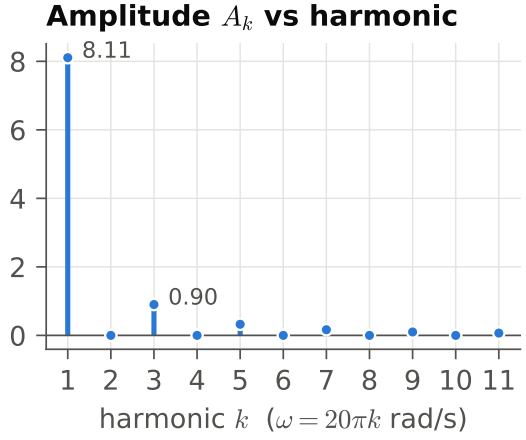
$$x(t) = \frac{A_0}{2} + \sum_{k=1}^{\infty} \Big[A_k \cos(k\omega_1 t) + B_k \sin(k\omega_1 t) \Big]$$

- $k\omega_1$ is the _____.
- $A_0/2$ is the _____.

2.1 Example: triangle wave (amplitude A, period T, even, zero mean)

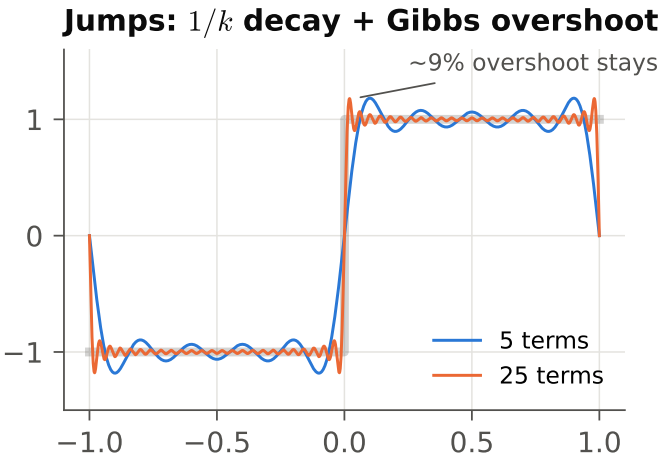
$x(t) = \sum_{k=1}^{\infty} \frac{8A}{\pi^2 (2k-1)^2} \cos\Big((2k-1) \frac{2\pi}{T} t \Big)$. With $A = 10$, $T = 0.1$ s:



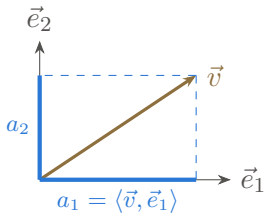


What do you notice about the harmonics?

Smoothness sets the decay. A continuous signal with corners (triangle) has $|A_k| \sim 1/k^2$. A signal with jumps (square wave) has only $\sim 1/k$, so it needs many more terms. At a jump the series converges to the midpoint and keeps a ≈9% overshoot (Gibbs phenomenon).

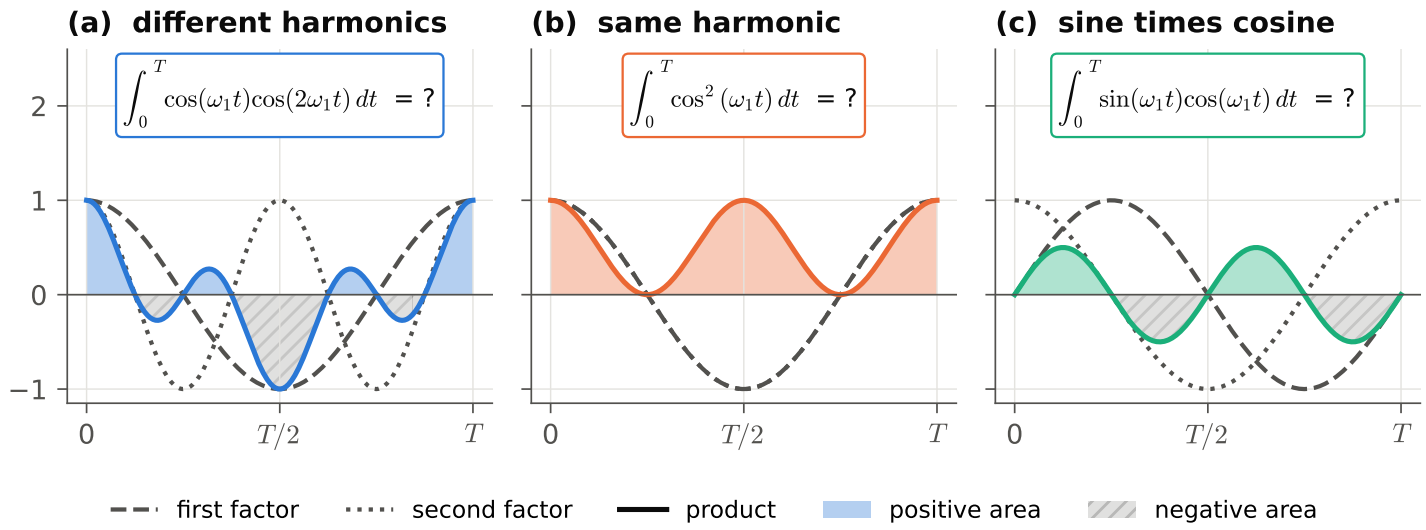


3 Geometric view: coefficients are projections



Vectors	Periodic functions
$\vec{v}$	$x(t)$
basis $\vec{e}_i$	$1,\ \cos k\omega_1 t,\ \sin k\omega_1 t$
$\langle \vec{v}, \vec{w} \rangle = \sum v_i w_i$	$\langle f, g \rangle = \int_0^T f\, g\, dt$
component a_i	coefficient A_k, B_k

To pull out one coefficient, say A_k, we multiply $x(t)$ by $\cos(k\omega_1 t)$ and integrate over one period. But $x(t)$ is itself a sum of harmonics (its Fourier series), so $x(t)\cos(k\omega_1 t)$ splits into one product per term of the series: $\cos(k\omega_1 t)$ times each harmonic in the series. Every one of those products is one of three kinds:

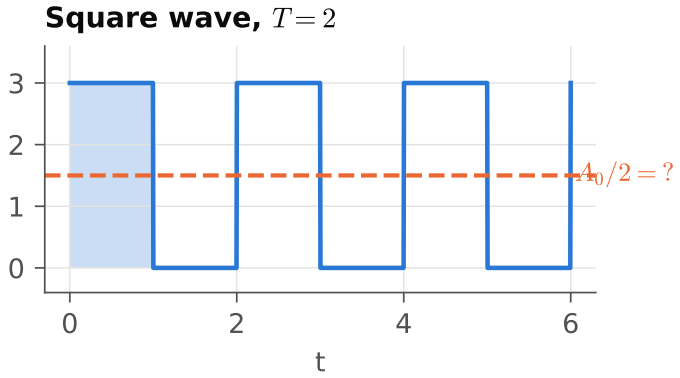


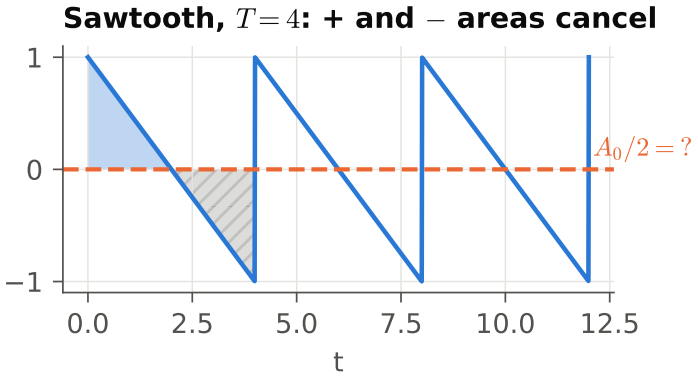
Compare the shaded areas above and below zero in each panel. Which integrals vanish, and which one does not?

Takeaway. So $\int_0^T x(t)\cos(k\omega_1 t)\,dt$ keeps only the k-th harmonic: it equals _____, which gives $A_k =$ _____.

4 Computing coefficients efficiently

4.1 DC term = average value

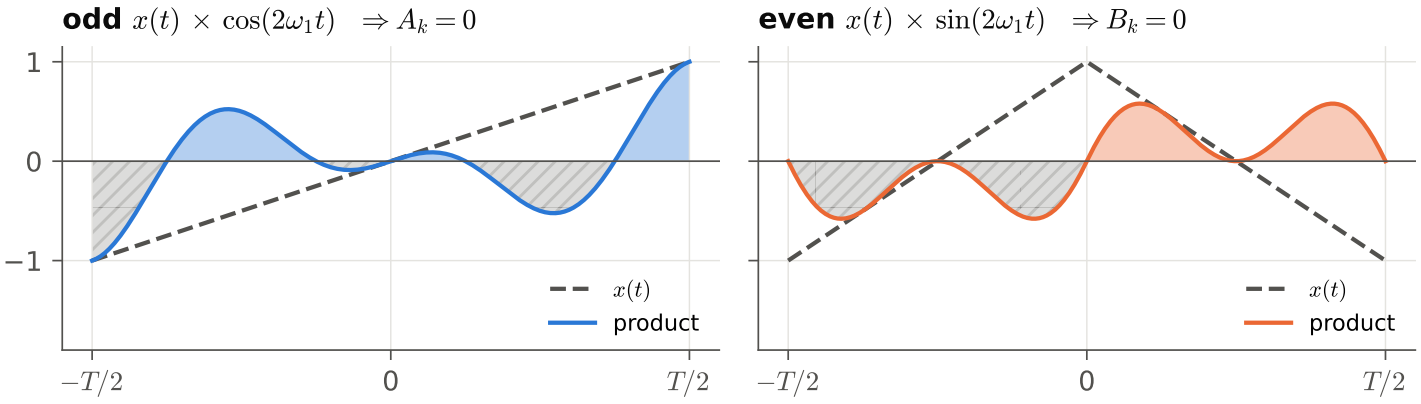




Square wave: $\frac{A_0}{2} = \frac{1}{T}\int_0^T x\,dt =$ _____ Sawtooth: $\frac{A_0}{2} =$ _____

Any full period works: $\int_{t_0}^{t_0+T}$ gives the same result for every t_0, because the integrand is T-periodic. Pick the limits that make the algebra easiest, often $[-T/2, T/2]$.

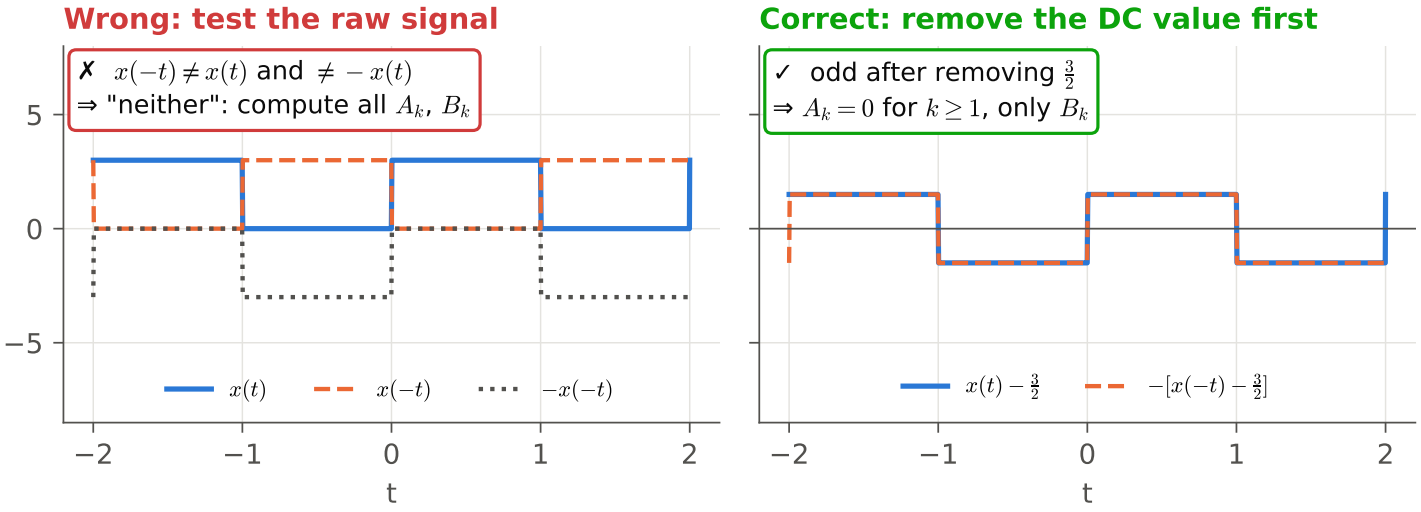
4.2 Symmetry shortcuts



odd × even = odd, and an odd function integrates to zero over $[-T/2, T/2]$.

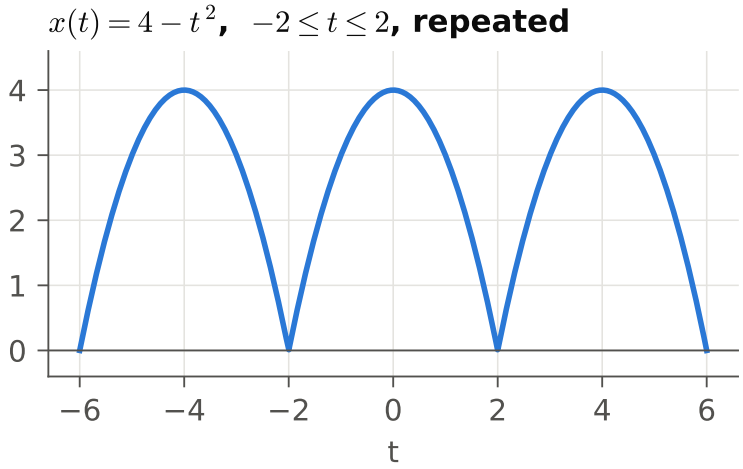
Symmetry	Test	Vanishing terms	Remaining integral
Odd	$x(-t) = -x(t)$		
Even	$x(-t) = x(t)$		
Half-wave	$x(t + \frac{T}{2}) = -x(t)$		only odd k survive

Watch out. Symmetry depends on where you put $t = 0$, and can appear only after you subtract the DC value. An even signal may still have a DC term. Most signals are neither odd nor even, so expect both A_k and B_k.



The 0–3 square wave from Section 4.1. Tested as is, it looks like “neither” (left). Subtract its mean $3/2$ first and it is odd (right), so only B_k terms survive, plus the DC term $3/2$.

5 Worked example 1: periodic parabola



Period $T =$ _____

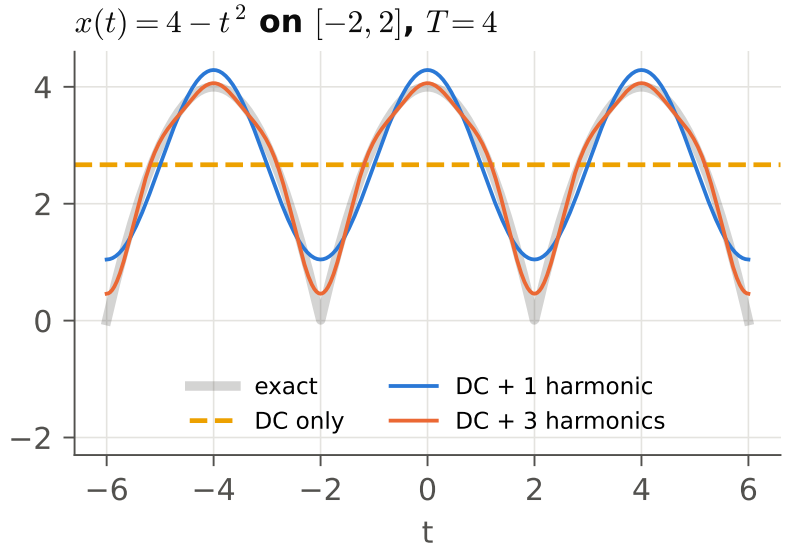
$\omega_1 =$ _____ rad/s

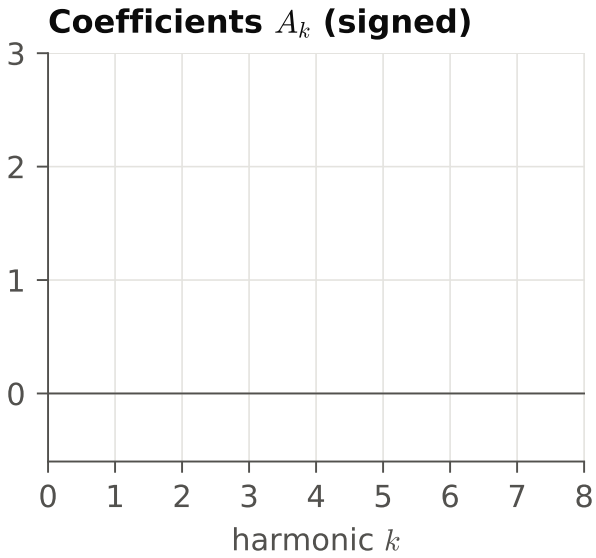
Symmetry: _____

DC term

Cosine coefficients ($\omega_1 = \pi/2$, so $k\omega_1 t = k\pi t/2$)

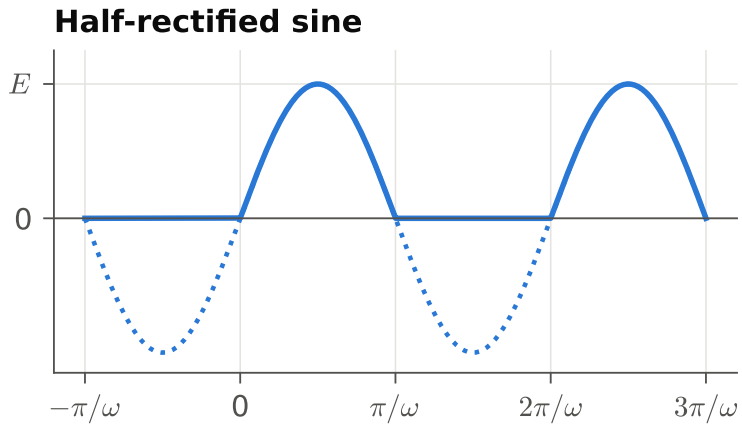
Series





Left: partial sums of the series. Right: plot $A_0/2$ and $A_1, \ldots, A_4$.

6 Worked example 2: half-rectified sine



$$x(t) = \begin{cases} E \sin \omega t & 0 < t < \pi/\omega \\ 0 & -\pi/\omega < t < 0 \end{cases}$$

$T =$ _____, $\omega_1 =$ _____

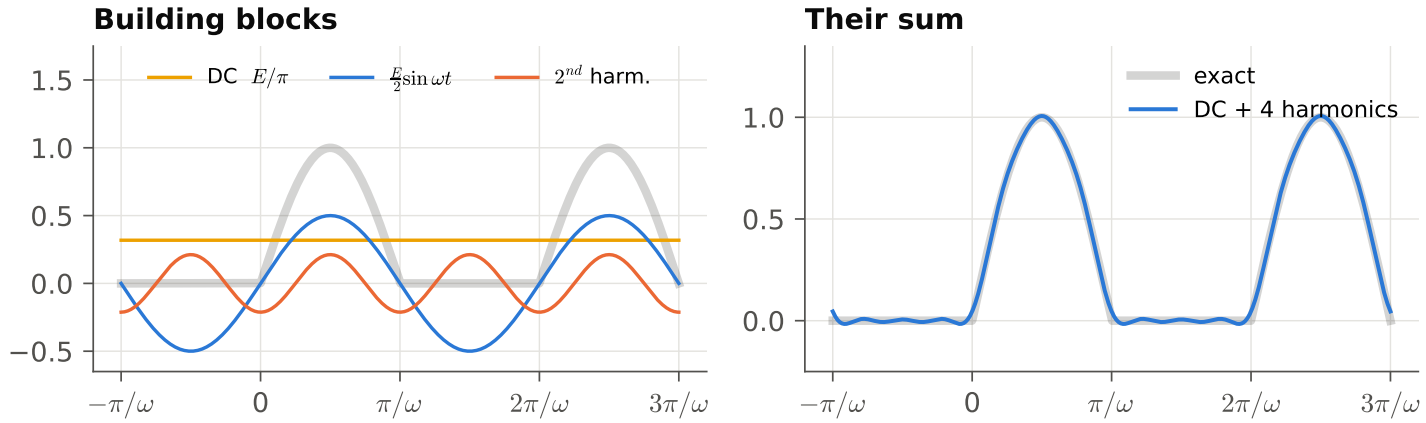
Odd? _____

Even? _____

DC component. Why does the integral stop at π/ω? _____

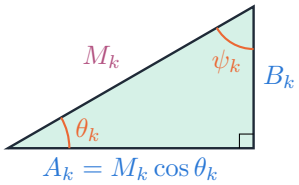
Sine and cosine coefficients (only $0 < t < \pi/\omega$ contributes)

Series



Left: the first terms of the series. Right: their sum compared with the exact signal.

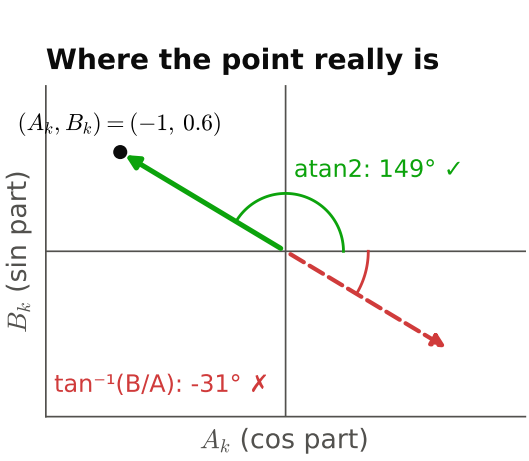
7 Magnitude–phase form (one-sided spectrum)

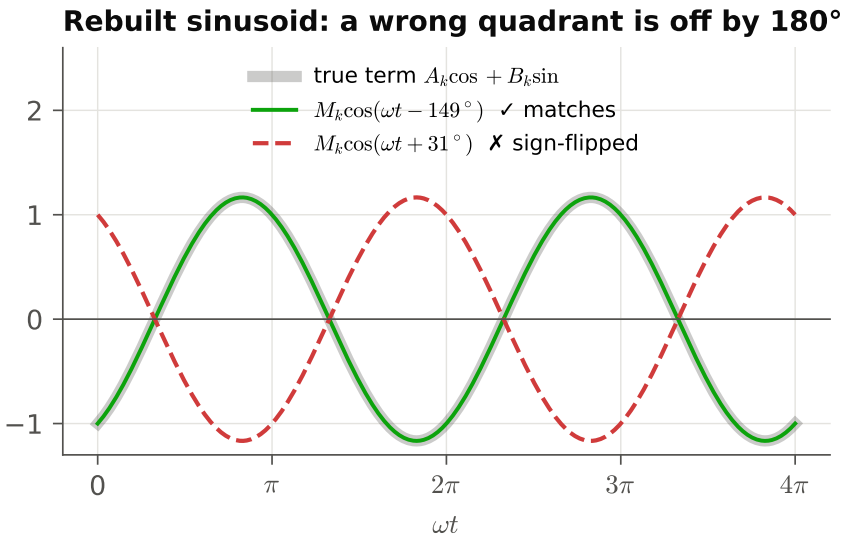


Expand $M_k \cos(k\omega_1 t - \theta_k)$ and match it with $A_k \cos + B_k \sin$:

$$x(t) = \frac{A_0}{2} + \sum_{k=1}^{\infty} M_k \cos(k\omega_1 t - \theta_k) = \frac{A_0}{2} + \sum_{k=1}^{\infty} M_k \sin(k\omega_1 t + \psi_k)$$

Watch out. $\tan^{-1}$ only returns angles in $(-\pi/2, \pi/2)$. Use the signs of A_k and B_k to pick the quadrant, i.e. $\theta_k = \operatorname{atan2}(B_k, A_k)$. For example, $A_k < 0$ and $B_k = 0$ gives $\theta_k = \pi$, not 0.

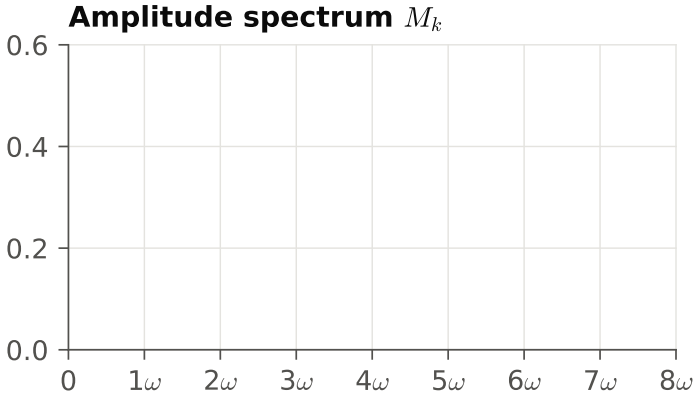


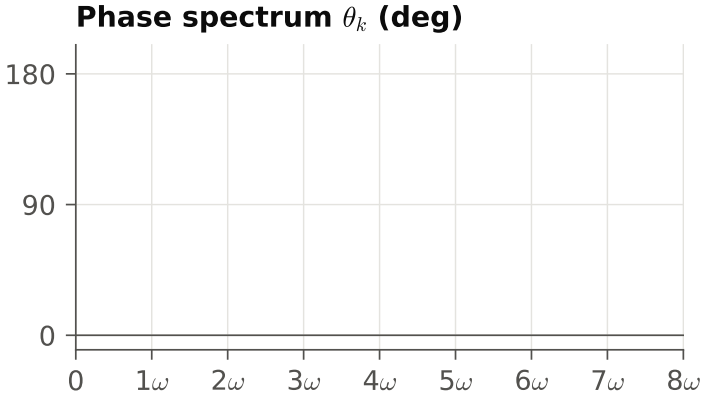


Example with $A_k = -1$, $B_k = 0.6$. A calculator's $\tan^{-1}(B_k/A_k)$ returns $-31°$, which points the opposite way (red). The rebuilt cosine is then the true term flipped upside down. $\operatorname{atan2}(B_k, A_k) = 149°$ (green) reproduces the true term exactly.

7.1 Half-rectified sine in magnitude–phase form

k	0	1	2	3	4
A_k	$A_0/2 = E/\pi$	0		0	
B_k	–	$E/2$	0	0	0
M_k				0	
θ_k	–			–	



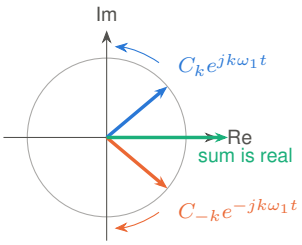


Plot the amplitude and phase spectra from the table.

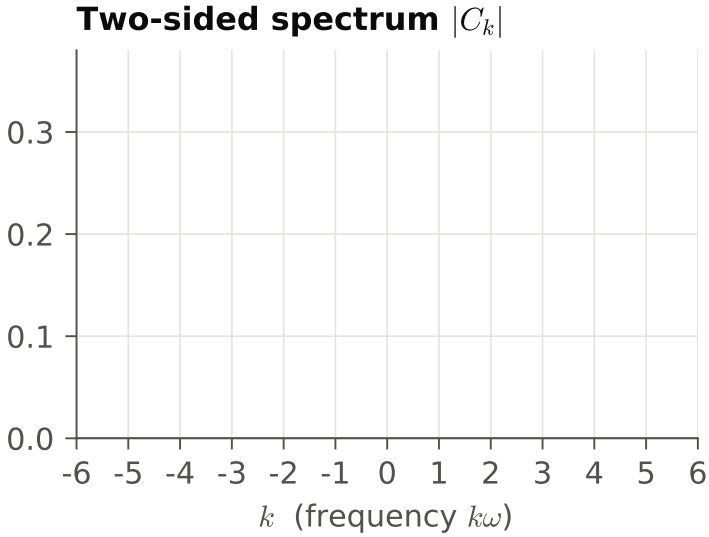
8 Complex exponential (Euler) form

Use $\cos\phi = \frac{1}{2}(e^{j\phi} + e^{-j\phi})$ and $\sin\phi = \frac{1}{2j}(e^{j\phi} - e^{-j\phi})$:

$$x(t) = \sum_{k=-\infty}^{\infty} C_k\, e^{jk\omega_1 t}, \qquad C_k = \frac{1}{T}\int_{-T/2}^{T/2} x(t)\, e^{-jk\omega_1 t}\, dt$$



Each real sinusoid is a pair of counter-rotating phasors, each with half the amplitude. “Negative frequency” is bookkeeping.



Half-rectified sine:

$C_0 =$ _____

$C_{\pm 1} =$ _____

$C_{\pm 2} =$ _____

The magnitude is symmetric in k and the phase is antisymmetric. The one-sided M_k is split in half between $\pm k$.

9 Summary: four equivalent forms

Form	Series	Coefficients
Sine–cosine	$\frac{A_0}{2} + \sum [A_k \cos k\omega_1 t + B_k \sin k\omega_1 t]$	compute by integration
Cosine series	$\frac{A_0}{2} + \sum M_k \cos(k\omega_1 t - \theta_k)$	
Sine series	$\frac{A_0}{2} + \sum M_k \sin(k\omega_1 t + \psi_k)$	
Complex	$\sum_{-\infty}^{\infty} C_k e^{jk\omega_1 t}$	

Back-conversion: $A_k = M_k \cos\theta_k =$ _____, $B_k = M_k \sin\theta_k =$ _____.

Exit question. $A_3 = -1$ and $B_3 = 1$. Find M_3, θ_3 and C_3.